Toward the Császár Polyhedron via Cell-Complex Enumeration

Step 1: Notation and the First Conditions

Notation

Let $P_i = (x_i, y_i, z_i) \in \mathbb{R}^3$ be the coordinate vector of the symbolic point i. Define the *bracket*

$$[abcd] \;:=\; \det\begin{pmatrix} 1 & x_a & y_a & z_a \\ 1 & x_b & y_b & z_b \\ 1 & x_c & y_c & z_c \\ 1 & x_d & y_d & z_d \end{pmatrix} \;=\; \det\bigl(P_b - P_a,\; P_c - P_a,\; P_d - P_a\bigr),$$

which is 6 times the signed volume of the tetrahedron $abcd$ and is alternating in a, b, c, d. With this,

$$abc^{\pm} \;:=\; \bigl\{\, p \in \mathbb{R}^3 \;:\; \pm[a, b, c, p] > 0 \,\bigr\}.$$

Condition 1: $3 \in 012^{+}$

This is just $[0, 1, 2, 3] > 0$. Three equivalent forms:

(a) Determinant.

$$\det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & x_3 & y_3 & z_3 \end{pmatrix} > 0 \quad\Longleftrightarrow\quad \det\begin{pmatrix} x_1 - x_0 & y_1 - y_0 & z_1 - z_0 \\ x_2 - x_0 & y_2 - y_0 & z_2 - z_0 \\ x_3 - x_0 & y_3 - y_0 & z_3 - z_0 \end{pmatrix} > 0$$

(b) Scalar triple product (cross-product antisymmetry).

$$(P_1 - P_0) \cdot \bigl((P_2 - P_0) \times (P_3 - P_0)\bigr) \;>\; 0.$$

This form privileges P_0 as a basepoint, hiding the S_4-antisymmetry on the four points. *Fully unparenthesized.* Distribute the differences (using $a{\times}a = 0$ and the cyclic identity $a{\cdot}(b{\times}c) = c{\cdot}(a{\times}b)$); the bracket reorganizes into the alternating sum over 3-element subsets of $\{0, 1, 2, 3\}$:

$$[0, 1, 2, 3] \;=\; P_1{\cdot}(P_2{\times}P_3) \;-\; P_0{\cdot}(P_2{\times}P_3) \;+\; P_0{\cdot}(P_1{\times}P_3) \;-\; P_0{\cdot}(P_1{\times}P_2).$$

Equivalently, with $T(i, j, k) := P_i \cdot (P_j \times P_k)$ (cyclically symmetric in i, j, k) and $\widehat{d}$ denoting the elements of $\{0, 1, 2, 3\} \setminus \{d\}$ in increasing order,

$$[0, 1, 2, 3] \;=\; \sum_{d=0}^{3} (-1)^d\, T\bigl(\widehat{d}\bigr).$$

The S_4-antisymmetry is now manifest — any swap of two points flips the bracket's sign.

(c) Polynomial in coordinates. With $\Delta_i := P_i - P_0$ for $i = 1, 2, 3$:

$$\Delta_{1,x}(\Delta_{2,y}\Delta_{3,z} - \Delta_{2,z}\Delta_{3,y}) + \Delta_{1,y}(\Delta_{2,z}\Delta_{3,x} - \Delta_{2,x}\Delta_{3,z}) + \Delta_{1,z}(\Delta_{2,x}\Delta_{3,y} - \Delta_{2,y}\Delta_{3,x}) \;>\; 0.$$

Expanding the Δ's as differences of original coordinates:

$$\begin{aligned} &(x_1{-}x_0)\big[(y_2{-}y_0)(z_3{-}z_0) - (z_2{-}z_0)(y_3{-}y_0)\big] \\ +\;&(y_1{-}y_0)\big[(z_2{-}z_0)(x_3{-}x_0) - (x_2{-}x_0)(z_3{-}z_0)\big] \\ +\;&(z_1{-}z_0)\big[(x_2{-}x_0)(y_3{-}y_0) - (y_2{-}y_0)(x_3{-}x_0)\big] \;>\; 0. \end{aligned}$$

Again the S_4-antisymmetry is hidden inside the differences. *Fully unparenthesized.* Distributing all subtractions yields the 24-term Levi-Civita expansion:

$$[0,1,2,3] \;=\; \sum_{(d,a,b,c)\in S_4} \operatorname{sgn}(d,a,b,c)\, x_a\, y_b\, z_c,$$

where (d, a, b, c) ranges over permutations of $(0, 1, 2, 3)$ and d is the row contributing the constant 1 from the bordered determinant (so d enters only through the sign). Grouped by missing index d:

$$\begin{aligned} d{=}0: &\quad +\,x_1y_2z_3 - x_1y_3z_2 - x_2y_1z_3 + x_2y_3z_1 + x_3y_1z_2 - x_3y_2z_1 \\ d{=}1: &\quad -\,x_0y_2z_3 + x_0y_3z_2 + x_2y_0z_3 - x_2y_3z_0 - x_3y_0z_2 + x_3y_2z_0 \\ d{=}2: &\quad +\,x_0y_1z_3 - x_0y_3z_1 - x_1y_0z_3 + x_1y_3z_0 + x_3y_0z_1 - x_3y_1z_0 \\ d{=}3: &\quad -\,x_0y_1z_2 + x_0y_2z_1 + x_1y_0z_2 - x_1y_2z_0 - x_2y_0z_1 + x_2y_1z_0 \end{aligned}$$

Every monomial uses each index at most once, and any swap of two indices flips the signs of all 24 terms in lock-step — the four-fold antisymmetry on $\{0, 1, 2, 3\}$ is entirely visible.

Condition 2: $012^+ \cap 013^+ \neq \emptyset$

Existence of p with $[0, 1, 2, p] > 0$ *and* $[0, 1, 3, p] > 0$.

(a) Determinant.

$$\exists\, p \in \mathbb{R}^3: \quad \det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & p_x & p_y & p_z \end{pmatrix} > 0 \;\wedge\; \det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_3 & y_3 & z_3 \\ 1 & p_x & p_y & p_z \end{pmatrix} > 0$$

(b) Scalar triple product.

$$\exists\, p \in \mathbb{R}^3: \quad (P_1{-}P_0)\cdot\big((P_2{-}P_0)\times(p{-}P_0)\big) > 0 \;\wedge\; (P_1{-}P_0)\cdot\big((P_3{-}P_0)\times(p{-}P_0)\big) > 0.$$

Fully unparenthesized. Each bracket cofactor-expands as in Condition 1(b), with p playing the role of the fourth point:

$$\begin{aligned} [0,1,2,p] &= P_1{\cdot}(P_2{\times}p) \;-\; P_0{\cdot}(P_2{\times}p) \;+\; P_0{\cdot}(P_1{\times}p) \;-\; P_0{\cdot}(P_1{\times}P_2), \\ [0,1,3,p] &= P_1{\cdot}(P_3{\times}p) \;-\; P_0{\cdot}(P_3{\times}p) \;+\; P_0{\cdot}(P_1{\times}p) \;-\; P_0{\cdot}(P_1{\times}P_3). \end{aligned}$$

The shared term $P_0{\cdot}(P_1{\times}p)$ reflects that both brackets share the edge $\overline{P_0P_1}$. Each is S_4-antisymmetric in its own four points ($\{0, 1, 2, p\}$ resp. $\{0, 1, 3, p\}$) and *affine* in p.

(c) Polynomial in coordinates. Substitute p for P_3 in the first inequality, and p for P_2 (keeping P_3 in the third slot) in the second. Writing $p = (x_p, y_p, z_p)$:

$$\exists\,(x_p, y_p, z_p) \in \mathbb{R}^3 :$$

$$\begin{aligned}
&(x_1{-}x_0)\big[(y_2{-}y_0)(z_p{-}z_0) - (z_2{-}z_0)(y_p{-}y_0)\big] \\
&+ (y_1{-}y_0)\big[(z_2{-}z_0)(x_p{-}x_0) - (x_2{-}x_0)(z_p{-}z_0)\big] \\
&+ (z_1{-}z_0)\big[(x_2{-}x_0)(y_p{-}y_0) - (y_2{-}y_0)(x_p{-}x_0)\big] \;>\; 0
\end{aligned}$$

AND

$$\begin{aligned}
&(x_1{-}x_0)\big[(y_3{-}y_0)(z_p{-}z_0) - (z_3{-}z_0)(y_p{-}y_0)\big] \\
&+ (y_1{-}y_0)\big[(z_3{-}z_0)(x_p{-}x_0) - (x_3{-}x_0)(z_p{-}z_0)\big] \\
&+ (z_1{-}z_0)\big[(x_3{-}x_0)(y_p{-}y_0) - (y_3{-}y_0)(x_p{-}x_0)\big] \;>\; 0.
\end{aligned}$$

Fully unparenthesized. Each bracket admits the 24-term Levi-Civita expansion of Condition 1(c). For $[0, 1, 2, p]$, treating p as the fourth point and grouping by missing index:

$$\begin{aligned}
d{=}0: &\quad + x_1y_2z_p - x_1y_pz_2 - x_2y_1z_p + x_2y_pz_1 + x_py_1z_2 - x_py_2z_1 \\
d{=}1: &\quad - x_0y_2z_p + x_0y_pz_2 + x_2y_0z_p - x_2y_pz_0 - x_py_0z_2 + x_py_2z_0 \\
d{=}2: &\quad + x_0y_1z_p - x_0y_pz_1 - x_1y_0z_p + x_1y_pz_0 + x_py_0z_1 - x_py_1z_0 \\
d{=}p: &\quad - x_0y_1z_2 + x_0y_2z_1 + x_1y_0z_2 - x_1y_2z_0 - x_2y_0z_1 + x_2y_1z_0.
\end{aligned}$$

$[0, 1, 3, p]$ is obtained by replacing every subscript 2 with 3 above. Two structural observations are now visible:

- Rows $d = 0, 1, 2$ contain all the p-dependence (linear in p); row $d = p$ is the constant $-\det(P_0, P_1, P_2)$ (resp. $-\det(P_0, P_1, P_3)$). So each bracket is affine in p, and the realizability question is whether two open halfspaces in p-space meet.
- The S_4-antisymmetry on the bracket's four points $\{0, 1, 2, p\}$ (or $\{0, 1, 3, p\}$) is manifest: every swap of two indices flips the signs of all 24 monomials simultaneously.

Realizability: $012^+ \cap 013^+$ exists, $012^- \cap 013^+ \cap 123^+ \cap 203^+$ does not

Notation conversion. The bracket $[a, b, c, d]$ is alternating, so $[2, 0, 3, p] = -[0, 2, 3, p]$, hence

$$203^+ \;=\; \{p : [2, 0, 3, p] > 0\} \;=\; \{p : [0, 2, 3, p] < 0\} \;=\; 023^-.$$

The cell in question is therefore $012^- \cap 013^+ \cap 023^- \cap 123^+$ in canonical (sorted-index) notation — the sign vector $(-, +, -, +)$ on the four faces $(012, 013, 023, 123)$ of tetrahedron 0123.

Key identity. For any $p \in \mathbb{R}^3$,

$$[0, 1, 2, p] \;-\; [0, 1, 3, p] \;+\; [0, 2, 3, p] \;-\; [1, 2, 3, p] \;=\; [0, 1, 2, 3]. \qquad (\star)$$

Proof. Since P_0, P_1, P_2, P_3 are affinely independent (because $[0, 1, 2, 3] \neq 0$), every $p \in \mathbb{R}^3$ has unique barycentric coordinates

$$p \;=\; \alpha_0P_0 + \alpha_1P_1 + \alpha_2P_2 + \alpha_3P_3, \qquad \alpha_0 + \alpha_1 + \alpha_2 + \alpha_3 \;=\; 1.$$

The lifted equality $(1,p) = \sum_i \alpha_i\,(1,P_i)$ holds in $\mathbb{R}^4$, so multilinearity of the bordered 4×4 determinant in row p gives, for distinct $a,b,c \in \{0,1,2,3\}$ with missing index d,

$$[a,b,c,p] \;=\; \sum_{i=0}^{3} \alpha_i\,[a,b,c,i] \;=\; \alpha_d\,[a,b,c,d],$$

since $[a,b,c,a] = [a,b,c,b] = [a,b,c,c] = 0$ (repeated row). Using $[0,1,3,2] = -[0,1,2,3]$, $[0,2,3,1] = +[0,1,2,3]$, $[1,2,3,0] = -[0,1,2,3]$ (reorder the missing index back into canonical position), and dividing by $[0,1,2,3]$:

$$\alpha_3 \;=\; \frac{[0,1,2,p]}{[0,1,2,3]}, \quad \alpha_2 \;=\; \frac{-[0,1,3,p]}{[0,1,2,3]}, \quad \alpha_1 \;=\; \frac{[0,2,3,p]}{[0,1,2,3]}, \quad \alpha_0 \;=\; \frac{-[1,2,3,p]}{[0,1,2,3]}.$$

Then $\alpha_0+\alpha_1+\alpha_2+\alpha_3 = 1$, multiplied through by $[0,1,2,3]$, is exactly $(\star)$. $\square$

Why $012^+ \cap 013^+$ is non-empty. Take the explicit witness

$$p^* \;:=\; 2P_3 - P_2 \qquad \text{(the reflection of } P_2 \text{ across } P_3\text{)}.$$

Its coefficients $(2,-1)$ sum to 1, so any affine function of p evaluated at p^* equals 2 times its value at P_3 minus its value at P_2. With $[0,1,2,P_2] = 0$ and $[0,1,2,P_3] = [0,1,2,3]$:

$$[0,1,2,\,p^*] \;=\; 2\,[0,1,2,3] \;-\; 0 \;=\; 2\,[0,1,2,3] \;>\; 0.$$

With $[0,1,3,P_2] = [0,1,3,2] = -[0,1,2,3]$ and $[0,1,3,P_3] = 0$:

$$[0,1,3,\,p^*] \;=\; 2\cdot 0 \;-\; (-[0,1,2,3]) \;=\; [0,1,2,3] \;>\; 0.$$

Both brackets are positive, so $p^* \in 012^+ \cap 013^+$. (Geometrically: planes 012 and 013 are distinct and share the line $\overline{P_0P_1}$, so they cut $\mathbb{R}^3$ into four open wedges, all non-empty; $012^+ \cap 013^+$ is one of them.)

Why $012^- \cap 013^+ \cap 023^- \cap 123^+$ is empty. Suppose for contradiction such a p exists. The four memberships translate to

$$[0,1,2,p] < 0, \quad [0,1,3,p] > 0, \quad [0,2,3,p] < 0, \quad [1,2,3,p] > 0.$$

Hence each summand on the left side of $(\star)$, with its prescribed sign, is *strictly negative*:

$$\underbrace{[0,1,2,p]}_{<0} \;\underbrace{-\,[0,1,3,p]}_{<0} \;\underbrace{+\,[0,2,3,p]}_{<0} \;\underbrace{-\,[1,2,3,p]}_{<0} \;<\; 0.$$

But by $(\star)$ this sum equals $[0,1,2,3]$, which is >0 by Condition 1. Contradiction. Hence $012^- \cap 013^+ \cap 023^- \cap 123^+ = \emptyset$. $\square$

Geometric meaning. In barycentric coordinates the four sign conditions read

$$\alpha_0 < 0, \quad \alpha_1 < 0, \quad \alpha_2 < 0, \quad \alpha_3 < 0 \qquad (\text{“}p \text{ on the wrong side of every face”}),$$

which is impossible because $\alpha_0+\alpha_1+\alpha_2+\alpha_3 = 1 > 0$. The four face-planes of 0123 in general position cut $\mathbb{R}^3$ into 15 open cells (the count $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}+\binom{4}{3}$), and the missing 16th sign vector is precisely $(012^-, 013^+, 023^-, 123^+)$. The asymmetry between the two cells of the question is now visible: $012^+ \cap 013^+$ slices $\mathbb{R}^3$ along only *two* planes (a wedge — always non-empty), while the missing 16th cell asks for a sign-coherent intersection across *all four* faces of a closed tetrahedron, which the affine relation $\sum \alpha_i = 1$ rules out.

Observations

- **Only signs matter.** The cell complex depends only on the *sign vector* in $\{+,-\}^{\binom{n}{3}}$. Coordinates are useful only as a realizability oracle: if we want a "without coordinates" enumeration, the coordinate forms are witnesses we consult only when the combinatorial sign rules cannot decide.
- **Condition 2 is automatic.** Planes 012 and 013 share the line $\overline{P_0P_1}$ and (since $P_3 \notin$ plane 012 by Condition 1) are distinct. They partition $\mathbb{R}^3 \setminus \overline{P_0P_1}$ into four open wedges, all non-empty. So $012^+ \cap 013^+ \neq \emptyset$ trivially. A concrete witness: any point $P_3 + \varepsilon\,\hat{n}_{013^+}$ with ε small.
- **Where it gets interesting.** Once we have all four planes $\{012, 013, 023, 123\}$ they cut $\mathbb{R}^3$ into $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}+\binom{4}{3} = 15$ regions. Of the $2^4 = 16$ a-priori sign vectors, exactly one is empty — the missing 16th cell. A clean way to spot it: the sign vector that says "p is on the *opposite* side of every face from the opposite vertex of tetrahedron 0123" is not realizable (no point sees every face from the wrong side).

Algorithm: deciding if a new plane $ij(n{+}1)$ slices an existing cell

We are now ready for the inductive step. Assume we have introduced $n+1$ named points $\{P_0, \ldots, P_n\}$ and constructed the cell complex of the $\binom{n+1}{3}$ planes through every triple. We pick a new point P_{n+1} inside one specific cell C^* of the current complex; this pins the signs of all 4-brackets involving $n{+}1$ (because membership of P_{n+1} in C^* is exactly $\sigma_{C^*}(\Pi) = \mathrm{sgn}([\Pi, n{+}1])$ for every existing plane Π). Then we must slice every existing cell with the $\binom{n+1}{2}$ new planes $\{\, ij(n{+}1) : 0 \le i < j \le n \,\}$. The basic question is: given a cell C and a new plane $\Pi = ij(n{+}1)$, does Π slice C (i.e. are both $C \cap \Pi^+$ and $C \cap \Pi^-$ non-empty)?

The algorithm below answers this question using the chirotope alone (no coordinates), and is correct in the general case. The conceptual primitive is the affine identity ($\star$) from earlier — now applied to *every* 4-tuple of named points, not just $\{0,1,2,3\}$.

State. We maintain

- the **chirotope**: $\mathrm{sgn}[a,b,c,d]$ for every 4-subset $\{a,b,c,d\} \subseteq \{0,\ldots,n{+}1\}$;
- the **cell complex** of the previous step: each open cell C stored as a sign vector $\sigma_C : \mathcal{H}_{\text{old}} \to \{+,-\}$ over the old planes $\mathcal{H}_{\text{old}} = \binom{\{0,\ldots,n\}}{3}$.

We do *not* need anything else — no coordinates, no magnitudes, no "rays at infinity."

The single sign-rule. For every 4-subset $\{a,b,c,d\}$ of named indices and any point p, the affine identity ($\star$) (proved earlier for $\{0,1,2,3\}$ but valid for any 4-tuple, by the same barycentric argument) reads

$$[a,b,c,p] \;-\; [a,b,d,p] \;+\; [a,c,d,p] \;-\; [b,c,d,p] \;=\; [a,b,c,d].$$

Taking signs and noting the four magnitudes on the LHS are positive, the LHS is forced to be strictly positive (regardless of magnitudes) iff the four signs $(\sigma(abc), -\sigma(abd), \sigma(acd), -\sigma(bcd))$ are all $+$, i.e. the cell sign pattern at the four "faces of tetrahedron $abcd$" is

$$\big(\sigma(abc), \sigma(abd), \sigma(acd), \sigma(bcd)\big) \;=\; (+,-,+,-).$$

Symmetrically the LHS is forced strictly negative iff the pattern is $(-,+,-,+)$. So:

Forbidden-pattern rule (per 4-tuple). A sign vector σ violates the chirotope at $\{a,b,c,d\}$ iff

$$\big(\sigma(abc),\sigma(abd),\sigma(acd),\sigma(bcd)\big) = \big(-,+,-,+\big)\cdot \operatorname{sgn}[a,b,c,d],$$

where the right-hand side means: when $\operatorname{sgn}[a,b,c,d]=+$ the forbidden pattern is $(-,+,-,+)$, and when $\operatorname{sgn}[a,b,c,d]=-$ the forbidden pattern is $(+,-,+,-)$. This is exactly the "missing 16th cell" pattern from the original section, applied to every tetrahedron formed by four named points.

Tope characterization. A sign vector σ over the planes $\binom{[N]}{3}$ corresponds to a non-empty open cell of the realized configuration iff it does not violate the forbidden-pattern rule at any 4-subset $\{a,b,c,d\}$ of named indices. (For a representable rank-4 chirotope — which we have, by construction — this is necessary and sufficient: it is the standard Grassmann–Plücker characterization of topes.)

Slice test. Given an old cell C with sign vector σ_C and a new plane $\Pi = ij(n{+}1)$, the slice question becomes a small constraint satisfaction problem. Try to extend σ_C to a full sign vector σ on $\binom{\{0,\dots,n+1\}}{3}$ by assigning a sign to each of the $\binom{n+1}{2}$ new planes $\big\{ab(n{+}1) : 0 \le a < b \le n\big\}$, subject to:

1. the pinned sign $\sigma(\Pi) = +$;
2. the forbidden-pattern rule for every 4-subset $\{a,b,c,n{+}1\}$ with $\{a,b,c\}\subset\{0,\dots,n\}$ (there are $\binom{n+1}{3}$ such subsets); each such constraint involves the known old-plane sign $\sigma_C(abc)$, the three new-plane signs $\sigma(ab(n{+}1)),\sigma(ac(n{+}1)),\sigma(bc(n{+}1))$, and the chirotope value $\operatorname{sgn}[a,b,c,n{+}1]$ (also known — it was pinned when P_{n+1} was placed in cell C^*).

4-tuples not containing $n{+}1$ are already satisfied by σ_C from the previous step, so they impose nothing new. If at least one extension exists, the half $C\cap\Pi^+$ is non-empty. Repeat with $\sigma(\Pi)=-$. The plane slices C iff both halves are non-empty.

Why this works. Each forbidden-pattern rule is a single "XOR"-style clause: it forbids exactly one assignment of the three new-plane signs (the unique completion that, together with $\sigma_C(abc)$ and the chirotope sign, would form the missing-16th-cell pattern of tetrahedron $\{a,b,c,n{+}1\}$). Out of $2^3=8$ possible sign triples on $\{ab(n{+}1),ac(n{+}1),bc(n{+}1)\}$, exactly 7 are admissible per clause, and the slice test is the conjunction over the $\binom{n+1}{3}$ clauses indexed by the choice of $\{a,b,c\}\subset\{0,\dots,n\}$.

Why this is also sufficient (and not merely necessary). Necessity is immediate from $(\star_{abcd})$: any realized p trivially satisfies every GP rule. The non-trivial direction is sufficiency — if σ violates no forbidden pattern, why must some p realize it?

The cleanest justification is via Farkas' lemma. The cell $C\cap\Pi^+$ is non-empty iff the strict-linear system

$$\big\{\ \sigma(abc)\,[a,b,c,p] > 0 : \text{all triples}\ \big\}$$

is feasible in $p\in\mathbb{R}^3$. By Farkas, infeasibility requires a non-negative combination $\sum \lambda_{abc}\,\sigma(abc)\,[a,b,c,p]$ that is identically ≤ 0 as a function of p. Since $[a,b,c,p]$ is affine, this means the linear part vanishes and the constant part is ≤ 0. The linear-part-vanishing condition is a non-negative-coefficient signed dependency among the plane-normals $\mathbf{n}_{abc}$.

Now: *every* such dependency comes (up to scaling) from the affine identity itself. Differentiating $(\star_{abcd})$ in p gives

$$\mathbf{n}_{abc} \;-\; \mathbf{n}_{abd} \;+\; \mathbf{n}_{acd} \;-\; \mathbf{n}_{bcd} \;=\; 0,$$

which is the unique alternating 4-term dependency of normals associated to the tetrahedron $\{a, b, c, d\}$. The non-negative Farkas combination matching this dependency forces the sign pattern σ to align as $(+, -, +, -)$ or $(-, +, -, +)$ on those four planes, and then the constant value of the combination is $\pm[a, b, c, d]$. The "≤ 0" condition on that constant is *exactly* the GP forbidden pattern.

So: Farkas-infeasibility $\iff$ some GP rule is violated. Contrapositive: GP-consistent $\iff$ realized. The algorithm is correct in the general case.

(Equivalently, in oriented-matroid language: we are dealing with a representable rank-4 oriented matroid, and Las Vergnas' tope characterization says topes are exactly the sign vectors satisfying the circuit sign-conditions, which for rank 4 are the $(\star_{abcd})$ rules. Mnëv's hardness applies to *abstract* chirotopes; we never face it, because we extend a realized configuration at every step.)

This is a CSP with $\binom{n+1}{2}$ Boolean variables and $\binom{n+1}{3}$ clauses — trivially solvable by backtracking for the values of n relevant here ($n \leq 6$ for everything up to the Császár polyhedron, giving 21 variables and 35 clauses).

Why this is not NP-complete in our setting. Realizability of an arbitrary abstract chirotope is $\exists\mathbb{R}$-complete (Mnëv), but that is not the question we ask: by construction we extend a chirotope that is already realized. For an already-realized rank-4 chirotope, the Grassmann–Plücker forbidden-pattern rule *characterizes* topes, so the slice CSP is a faithful chirotope-only test. The CSP itself has small fixed arity (each clause touches only three new-plane variables), so it lies far below the worst-case complexity wall.

Per-step complexity. For step $n \to n{+}1$: $\binom{n+1}{2}$ new planes, each tested against every old cell. The number of old cells is bounded by Zaslavsky's count, $O(n^9)$. Each individual slice test is a CSP of fixed structure on $\binom{n+1}{2}$ variables and $\binom{n+1}{3}$ clauses, each of arity 3; backtracking solves each instance in time polynomial in n. The whole step is therefore polynomial in n (dominated by the cell count), and entirely chirotope-based.

Segment-triangle piercing in bracket form

A natural geometric primitive (and a check we will need later, e.g. for "does this Császár candidate actually embed without crossings?"): given two segments and a triangle made of named points, does the segment $\overline{P_iP_j}$ pierce the (open) triangle $\triangle P_kP_lP_m$? The answer is a four-line test in the chirotope, no coordinates needed.

Two conditions. Assume the five points are in general position (no 4 coplanar, so all five 4-brackets on $\{i, j, k, l, m\}$ are nonzero). The segment pierces the triangle iff both:

1. *The line through P_i, P_j pierces the triangle's interior*, i.e. the three signed volumes of segment-against-edge tetrahedra all agree:

$$\operatorname{sgn}[i, j, k, l] \;=\; \operatorname{sgn}[i, j, l, m] \;=\; \operatorname{sgn}[i, j, m, k] \;\in\; \{+, -\}.$$

Each $[i, j, \cdot, \cdot]$ is, up to sign, the signed volume of the tetrahedron formed by the segment and one edge of the triangle. Their common sign records the directed orientation of $\overline{P_iP_j}$ "going around" the triangle. If any one differs, the line misses or grazes that edge.

2. *The two endpoints P_i, P_j lie on opposite sides of the plane through klm*, so the crossing point of the line with the plane falls inside the segment:

$$\operatorname{sgn}[k, l, m, i] \neq \operatorname{sgn}[k, l, m, j].$$

So the piercing test reads off five 4-bracket signs on $\{i, j, k, l, m\}$ and checks two equalities. The five brackets are tied by the GP identity $\sum_q (-1)^q [\widehat{q}] = 0$, so any one is determined up to sign by the other four; in practice we just take whichever signs the chirotope already has.

Two-segment intersection. As an aside: two segments $\overline{P_iP_j}$ and $\overline{P_kP_l}$ in $\mathbb{R}^3$ can meet inside their interiors only if they are coplanar, which requires $[i, j, k, l] = 0$ — a non-generic condition. So in the generic chirotope two named segments share at most a common endpoint, and $\operatorname{sgn}[i, j, k, l]$ records which of the two segments "goes above" the other (the sign of the lift).

Redundant halfspaces in cell descriptions

A cell of our arrangement is described by signs on *every* plane, but only some of these signs are *essential*: dropping a non-essential one leaves the cell unchanged. A non-essential constraint is *redundant*; the planes whose constraints are essential are the *facets* of the cell.

The tetrahedron example. In the four-plane arrangement of tetrahedron 0123, the cell "past P_3" (the open cone past vertex P_3, bounded below by the three faces meeting at P_3) has sign vector

$$\big(\sigma(012), \sigma(013), \sigma(023), \sigma(123)\big) = (+, +, -, +).$$

It really has only three facets — the three faces $013, 023, 123$ that meet at P_3. The fourth sign $\sigma(012) = +$ is redundant: the cell happens to lie on the same side of 012 as the tetrahedron's interior, but the plane 012 is not a face of the cell. Equivalently, flipping $\sigma(012)$ to $-$ would give the sign vector $(-, +, -, +)$, which is precisely the missing 16th cell — empty. So this constraint cannot be the binding one.

The chirotope-only redundancy test. The previous paragraph generalizes: *the constraint σ_j on plane $\Pi_j = abc$ is redundant for a cell C iff flipping it produces an empty cell, iff some forbidden-pattern rule is triggered after the flip.* Concretely,

> $\sigma(abc)$ is redundant for C iff there is some $d \notin \{a, b, c\}$ such that, after flipping $\sigma(abc)$, the four signs $\big(\sigma(abc), \sigma(abd), \sigma(acd), \sigma(bcd)\big)$ form the forbidden pattern $(-, +, -, +) \cdot \operatorname{sgn}[a, b, c, d]$ of the 4-tuple $\{a, b, c, d\}$.

This is a pure chirotope check, with $O(n)$ work per plane (one 4-tuple per choice of d). It tells us, for each cell, exactly which sign-constraints are facets and which are mere consequences of the rest.

Speeding up cell splitting. The same machinery serves as a pre-filter for the slice test. To decide whether a new plane Π slices cell C, we previously had to ask the LP about both extensions σ^+, σ^-. With the chirotope filter:

1. Run the forbidden-pattern check on σ^+ and on σ^- in $O(n)$ time, considering only the 4-tuples that involve Π's three indices.
2. If both σ^+ and σ^- are forbidden-free: Π slices C, do the actual split.
3. If exactly one is forbidden: the cell lies entirely on the other side; attach that sign to C and move on without an LP.
4. Both being forbidden cannot happen for an existing cell — C has a witness which already realizes one extension.

In our $n = 7$ run with ~ 3000 cells, the new-plane stage at $n = 8$ would ask $\sim 3000 \times 21$ slice questions; the filter answers most of them without any LP, since for most cells most new planes are redundant.

Storing only facets. As a representational optimization, each cell can be stored as the dict of its *facet* signs (with the rest re-derived from the chirotope on demand). With this convention, the interior of tetrahedron 0123 lists all four signs (four facets); the past-P_3 cone lists only three; and the forbidden 16th cell, if it were realized, would list zero — which is precisely the way to recognize that it is empty.